Breast Cancer Classification and Segmentation on Ultrasound Images: A Leakage-Free Benchmark, an Improved Segmenter, and a Rigorously Evaluated Classifier

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Abstract

Breast ultrasound (BUS) is safe and cheap but hard to interpret, and its main public benchmark, BUSI, is small, imbalanced and full of near-duplicate images. We study lesion segmentation and image classification on BUSI under one cleaned, *leakage-free* protocol. Perceptual hashing shows that 36% of BUSI lies in near-duplicate clusters and that an exact cross-class duplicate exists, so we split at the cluster level. Under this protocol we reproduce thirteen published models and add two of our own. A controlled ablation shows an ImageNet-pretrained encoder alone raises Dice from 0.648 to 0.765 and accounts for essentially all of our segmenter's gain, while a Focal-Tversky loss, test-time augmentation and attention gates do not improve overlap. A controlled Stage-1 substitution shows the dual-stage pipeline is limited by its *classifier*, not its segmenter. Evaluated rigorously (out-of-fold training masks, five seeds, bootstrap 95% CIs), hand-crafted shape/radiomics features give no reliable gain and only ensembling survives, reaching macro-F1 0.875 (95% CI $[0.80, 0.94]$). Our reproductions fall below their published numbers, which we attribute to removing leakage. This is a decision-support study, not autonomous diagnosis.

Keywords—breast ultrasound; BUSI; medical image segmentation; image classification; U-Net; transfer learning; data leakage; near-duplicate detection; ablation study; bootstrap confidence interval; computer-aided diagnosis.

1 Introduction

Breast cancer is a leading cause of cancer death in women and early detection is the most effective way to reduce mortality [19]. Breast ultrasound is safe, cheap and radiation-free, and works well on dense tissue where mammography loses sensitivity, but it is hard to read: speckle noise, low contrast, acoustic shadowing and blurred boundaries make manual assessment slow and operator-dependent [18]. *Classification* answers *what* a scan shows (normal / benign / malignant); *segmentation* answers *where* the lesion is, supporting interpretability, measurement, and a classifier focused on the lesion rather than surrounding clutter. Automating both would let a computer-aided diagnosis system triage screening studies and give the radiologist a second reading, which is the practical application we target.

Following the Project 047 brief we build and evaluate a system on the BUSI dataset [1] for both tasks. Our central concern is *trustworthy comparison*: BUSI is small, imbalanced and, as we show, riddled with near-duplicates, so a careless split inflates scores substantially. We therefore build one cleaned, leakage-free protocol and put every model through it. Four questions organise the work. **RQ1:** how do established architectures perform under it, versus their published numbers? **RQ2:** which design decisions actually matter on data this small? **RQ3:** does better segmentation improve downstream classification? **RQ4:** are apparent gains stable across seeds and resampling?

Contributions. (i) A near-duplicate analysis of BUSI and a cluster-grouped, class-stratified leakage-free split. (ii) A unified reproduction of *thirteen* published models – eight segmenters, three classification backbones, one dual-stage pipeline, one multi-task network – under that single protocol. (iii) An improved pretrained-encoder segmenter with a controlled ablation isolating each ingredient. (iv) An improved dual-stage classifier with a controlled Stage-1 substitution and a ground-truth-ROI upper-bound diagnostic. (v) A rigorous evaluation protocol (out-of-fold masks, multi-seed ensembling, bootstrap CIs) that separates real improvements from noise.

2 Related Work

Classical BUS CAD chained pre-processing, segmentation, feature extraction and classification; the surveys of Cheng et al. [19] and Xian et al. [18] recur to the same obstacles that persist today. BUSI [1] is the de-facto benchmark because it supplies both labels and pixel masks. A documented data-quality issue – repeated and near-duplicate frames, some with conflicting labels – has been raised for it [21]; we quantify this directly and design our evaluation around it.

Classification. ResNet [4] made very deep networks trainable via identity shortcuts; DenseNet [5] promotes feature reuse, attractive on small data; EfficientNet [6] compound-scales depth/width/resolution and transfers well to medical images in its small variants. Applied to BUSI, Latha et al. [12] report $\approx$ 99% accuracy with EfficientNet-B7; such near-perfect numbers on so small a dataset invite suspicion of leakage, which directly motivates our protocol.

Segmentation. U-Net [3] is the foundational encoder–decoder with skips and trains well from few images; UNet++ [8] adds nested skips. Yap et al. [2] first applied deep learning to BUS lesion detection, comparing a patch LeNet, a U-Net and a transfer-learned FCN-AlexNet (best). Attention U-Net [9] gates the skips so the network learns where to look; DeepLabV3+ [7] adds atrous separable convolutions and ASPP for multi-scale context; nnU-Net [15] self-configures the whole pipeline and remains a strong general baseline. Two designs target BUS specifically: AAU-Net [10] replaces convolutions with a hybrid adaptive attention module for blurred boundaries, and CResU-Net [11] schedules specialised residual/dense blocks and reports the strongest BUSI Dice in this literature. Foundation segmenters (SAM [16], MedSAM [17]) need prompts or adaptation, so we treat them as context, not baselines. For the combined task, Bruno et al. [13] propose a *sequential* pipeline (segment, crop the ROI, classify it) whereas Lu et al. [14] use *joint* multi-task learning with a shared encoder; these two lines frame our design space.

Limitations of existing work. Three recur. Reported BUSI scores use different splits and pre-processing, so they are not comparable across papers; most studies report a *single* run with no seeds or confidence intervals, on a test set far too small to resolve the differences they claim; and none de-duplicates, so near-duplicates leak across a naive split and inflate every number. Our protocol addresses each in turn, which is what makes the reproductions below comparable to one another even though they are lower than published values.

3 Methods

3.1 Reproduced models

All models share the same split, letterbox pre-processing, augmentation and metric code from `Models/common/`, so differences reflect architecture and recipe, not data handling. For every model we wrote the head/decoder and the training/evaluation loop ourselves; pretrained backbones are torchvision or `segmentation-models-pytorch` components. We reproduce the brief's three required papers – a from-scratch **U-Net** [3], Yap's **patch LeNet** and **FCN-AlexNet** [2] – plus **Attention U-Net** [9] (identical recipe to our U-Net, hence a controlled test of the attention gate), **DeepLabV3+** [7] (pretrained ResNet-50 + ASPP), **nnU-Net** [15] (blueprint 2D U-Net, deep supervision), **AAU-Net** [10] (HAAM module from scratch) and **CResU-Net** [11] (pure Dice loss as published). For classification we fine-tune the *smallest* variant of each family – **ResNet-18**, **DenseNet-121**, **EfficientNet-B0** – with a fresh three-way head and weighted cross-entropy, because BUSI is small. For the combined task we reproduce Bruno's **dual-stage** pipeline and Lu's **MTL-OCA** multi-task network.

3.2 Our two proposed models

Improved segmenter. The only reproduced segmenter using transfer learning (DeepLabV3+) clearly out-scored the from-scratch ones, so we hypothesise that *transfer learning is the decisive factor* on BUSI. We therefore build a U-Net whose encoder is an ImageNet-pretrained **EfficientNet-B4**, keeping the U-Net decoder and skips (20.2 M parameters). On top we test two EDA-driven ideas: a **Focal-Tversky + BCE** loss and **TTA + post-processing**. The Tversky index $T = \mathrm{TP}/(\mathrm{TP} + \alpha\,\mathrm{FP} + \beta\,\mathrm{FN})$ weights false positives and negatives asymmetrically (soft counts, per image), and the focal variant is $\mathcal{L}_{\mathrm{FT}} = (1-T)^{\gamma}$ [20]. We train with $\mathcal{L} = \lambda\mathcal{L}_{\mathrm{BCE}} + (1-\lambda)\mathcal{L}_{\mathrm{FT}}$, $\lambda = 0.5$, $(\alpha,\beta,\gamma) = (0.3, 0.7, \frac{4}{3})$ – the published defaults, *not* tuned on our data, so the ablation stays honest; since $\alpha{=}\beta{=}0.5, \gamma{=}1$ recovers Dice exactly, this strictly generalises the baseline loss. At inference TTA averages the probability maps of the image, its horizontal flip and two rescalings; the result is thresholded at 0.5, closed with a 5×5 ellipse and reduced to its largest connected component.

Improved dual-stage pipeline. We make three changes to Bruno's pipeline and evaluate them rigorously. *(i) Stronger Stage-1:* we substitute our segmenter and keep the ROI coupling and Stage-2 classifier byte-for-byte identical, so any change is attributable to segmentation alone. The coupling is fixed throughout: largest 8-connected component, bounding box dilated by 15% per side, rejected below 16 px, falling back to the whole frame if the mask is empty, resized to $256{\times}256$. *(ii) Strengthened Stage-2:* the original trains on clean ground-truth ROI crops but tests on imperfect predicted crops – a distribution mismatch. We train on *both*, select on predicted-ROI validation macro-F1, and add flip TTA. *(iii) Feature fusion:* we optionally concatenate 6 shape descriptors (area ratio, circularity, solidity, extent, bbox aspect, eccentricity) or a 14-D radiomics vector (+3 intensity, +5 GLCM texture) to the pooled CNN embedding, always computed from the *predicted* mask so the classifier never sees geometry it would lack at deployment.

4 Experiments

Dataset. We use the Breast Ultrasound Images (BUSI) dataset [1], collected in 2018 at Baheya Hospital, Cairo, from 600 female patients aged 25–75. It holds 780 grayscale PNG scans (average $\approx 500 \times 500$ px), each with a label in {normal, benign, malignant} and one or more pixel-level lesion masks; normal scans carry an empty mask. Source: `huggingface.co/datasets/gymprathap/Breast-Cancer-Ultrasound-Images-Dataset`.

Key EDA findings. The classes are imbalanced – benign 437 (56%), malignant 210 (27%), normal 133 (17%) – so we report *macro*-averaged metrics rather than accuracy and weight the loss by inverse class frequency. Sizes vary (639 distinct resolutions), so we resize *with padding* to $256{\times}256$ to preserve the discriminative lesion aspect ratio. Malignant lesions occupy a far larger area ratio (median $\approx 12\%$ vs. $\approx 4\%$) and many benign lesions are tiny, creating a severe pixel-level foreground imbalance that favours an overlap-aware loss; they are also less circular and solid and more eccentric, motivating both attention-based segmenters and the shape-feature fusion of Section 3.2. Lesions concentrate near the image centre, so we augment with small translations.

The decisive finding is a data-integrity one. MD5 hashing reveals an *exact cross-class duplicate*: `benign(433).png` and `malignant(145).png` are byte-identical but carry different labels. Perceptual hashing (Hamming ≤ 5) finds 186 near-duplicate pairs, 10 of them crossing class boundaries, forming 124 clusters that cover 277 images – 36% of BUSI. Simulating naive random 70/30 splits, on the order of 57 clusters end up with copies on *both* sides, so the test set contains images the model effectively trained on. Our remedy: drop the conflicting duplicate, union-merge multi-mask images, and split at the *cluster* level while stratifying by class.

Split. One cluster-grouped, class-stratified split is built once and cached, so every training script reads the same file: clusters (not images) are shuffled with seed 42 within each class and allocated 70/15/15. Segmentation uses the 645 lesion images, split 445/102/98 over 532 clusters; classification uses all 778 retained images, split 540/120/118 over 626 clusters; the dual-stage pipeline is scored on the 98-image two-class test subset. A programmatic check confirms **zero** clusters span more than one split – the assertion that makes the benchmark leakage-free. Augmentation (horizontal flip p=0.5; affine scale $[0.9, 1.1]$, translation $\pm 5\%$, rotation $\pm 15°$; brightness/contrast ± 0.2, p=0.3) is applied to the training split only. We deliberately omit vertical flips: an ultrasound frame has a fixed top-down geometry.

Training. Each model follows its source paper's recipe where practical, rather than a single forced recipe that would misrepresent the reproductions; all inputs are 256×256. Segmenters use Adam/AdamW at 10^{-3}–10^{-4} for 60–120 epochs (batch 8), except nnU-Net (200 epochs, SGD-Nesterov 0.99, poly$^{0.9}$) and MTL-OCA (300). Classifiers use batch 16, 20–60 epochs, weighted cross-entropy. Our improved segmenter uses AdamW (10^{-3}, cosine, 60 epochs), the improved Stage-2 Adam (10^{-4}, cosine, 60). Masks are thresholded at a fixed 0.5; Patch-LeNet's detection threshold was tuned on validation only (0.8). All selection uses train/validation only.

Metrics. Segmentation: per-image Dice and IoU (primary, averaged over images so large lesions do not dominate), pixel precision/recall and Yap's detection metrics (true-positive fraction, FP per image, F-measure). Classification: macro-F1 (primary), balanced accuracy, accuracy, per-class P/R/F1 and macro one-vs-rest AUC. Macro-F1 is preferred because accuracy can look high while the minority malignant class is handled badly.

Statistical protocol. For the final pipeline we go beyond single runs. Each configuration is trained with *five seeds*; we report the per-seed mean±std and a five-model probability-averaged *ensemble*, plus a percentile *bootstrap* 95% CI (B = 2000 resamples of the 98 test images) for test-set uncertainty. Predicted *training* masks are generated *out-of-fold* by five-fold cross-fitting, so each training image's mask comes from a segmenter that never saw it – a correctness measure, not a way to raise the score. Overlapping CIs are treated as evidence that a difference is *not* resolved. Everything ran on one RTX 5060 laptop GPU (PyTorch 2.10, CUDA 12.8), so wall-clock times are comparable; total compute is about six GPU-hours.

Table 1: Segmentation on the leakage-free test set (98 images), by Dice. σ = per-image Dice std; P/R = pixel precision/recall; F_{det} = Yap detection F-measure; "Paper" = published BUSI Dice where available.

Model	Dice	σ	IoU	P	R	F_{det}	M	Paper
Improved U-Net (ours)	**.760**	.307	**.688**	.765	.790	**.854**	20.2	–
DeepLabV3+ [7]	.739	.290	.652	.765	.753	.810	26.7	–
AAU-Net [10]	.730	**.284**	.638	.739	.759	.723	42.7	.775
nnU-Net [15]	.717	.299	.627	.754	.719	.781	8.5	–
CResU-Net [11]	.693	.305	.600	.759	.697	.772	33.5	.829
U-Net [3]	.648	.362	.570	**.775**	.637	.779	31.0	–
Attn. U-Net [9]	.629	.345	.540	.701	.638	.631	31.4	–
MTL-OCA [14] (seg)	.612	.332	.515	.737	.598	.539	8.2	–
FCN-AlexNet [2]	.582	.310	.471	.648	.647	.804	28.7	–
Patch-LeNet [2]	.285	.268	.198	.641	.223	.598	0.4	–

Table 2: Controlled ablation of the improved segmenter. Each row adds one component; split, pre-processing, augmentation and schedule are held fixed.

Configuration	Dice	Δ	σ	Pix. P	Pix. R	FP/img
From-scratch U-Net (Dice+BCE)	.648	—	.362	.775	.637	.296
+ pretrained EffB4 enc.	**.765**	+.117	.285	**.799**	.763	.214
+ Focal-Tversky loss	.760	−.005	**.282**	.775	**.800**	.225
+ TTA + post-proc. (full)	.760	±.000	.307	.765	.790	**.163**

5 Results

5.1 Segmentation

Our improved segmenter is best (Dice **0.760**, IoU 0.688), ahead of DeepLabV3+ (0.739) and both BUS-specific SOTA reproductions (Table 1). Three findings follow. First, our reproductions score *below* their published numbers (AAU-Net 0.730 vs. 0.775; CResU-Net 0.693 vs. 0.829), consistent with removing the near-duplicate leakage the original protocols did not. Second, nnU-Net (0.717), the general "strong baseline", does *not* dominate here – the transfer-learning models beat it. Third, the controlled pair Attention U-Net (0.629) vs. U-Net (0.648), which share an identical recipe, shows the attention gate does not help on this data. Almost every model is precision-dominant, so the shared failure mode is under-segmentation; ours is the only one whose recall exceeds its precision, the clinically preferable direction and the intended effect of the FN-weighted loss.

Ablation (Table 2). The result is unambiguous: the pretrained encoder is responsible for essentially the *entire* gain, raising Dice from 0.648 to 0.765 (+0.117) and cutting the Dice std from 0.362 to 0.285 – i.e. far fewer completely-missed lesions. It is also the *cheapest* model to train (7.9 vs. 58.5 min for the U-Net it replaces), because pretraining converges in far fewer epochs. Focal-Tversky does *not* raise Dice (−0.005); consistent with its design it trades 2.4 points of precision for 3.7 of recall and recovers two more lesions. Dice, symmetric in FP and FN, is almost blind to that reallocation – which is why we report the breakdown rather than resting on the headline. TTA leaves Dice unchanged but cuts false positives by a quarter. We therefore explicitly *do not* claim Focal-Tversky or TTA improve overlap; their effects are on recall and output cleanliness.

5.2 Classification

EfficientNet-B0 is best (macro-F1 **0.828**, balanced accuracy 0.852, AUC 0.969), ahead of ResNet-18 (0.810)

Table 3: Improved dual-stage pipeline, two-class 98-image test set, evaluated with out-of-fold training masks, five seeds and a bootstrap 95% CI ($B = 2000$). All three intervals overlap; only ensembling reliably helps.

Configuration	Ens. F1	Per-seed	95% CI	Malig. R
OOF, no hand features	**.875**	.847±.011	[.800, .936]	**.903**
OOF + shape (6)	**.875**	.861±.016	[.800, .936]	**.903**
OOF + radiomics (14)	.863	.862±.013	[.785, .929]	.871

and DenseNet-121 (0.793); all three pretrained backbones clearly beat MTL-OCA (0.720). Per-class F1 for the best model is normal 0.766, benign 0.884, malignant 0.833, with malignant recall 0.807 – this per-class view, not the overall accuracy of 0.848, is the operative summary for an imbalanced clinical task. The ranking also hides a clinically relevant detail: ResNet-18 is *more sensitive* on malignant ($R = 0.839$, one more of 31 caught) though less precise, so it is arguably the better operating point despite ranking second. MTL-OCA underperforms on *both* tasks (Dice 0.612, macro-F1 0.720): sharing one backbone across tasks is, on data this small, worse than a specialised network per task.

5.3 Dual-stage pipeline

Controlled Stage-1 substitution (RQ3). Substituting our stronger segmenter raises Stage-1 Dice from 0.739 to 0.760 but moves single-run end-to-end macro-F1 only from 0.804 to 0.807 – about 0.14 points of F1 per point of Dice. The GT-ROI diagnostic explains why: feeding the *same* classifier perfect crops reaches only 0.865–0.884, so the score recoverable by perfect segmentation is 0.06–0.08 and does not shrink as Stage-1 improves. Concretely, the pipeline misclassifies 16 of 98 images and the oracle misclassifies 10: perfect masks repair six errors and leave ten standing. A mask with Dice 0.76 already crops a useful ROI, so the system is *classifier-bound*, not segmentation-bound.

Rigorous evaluation (RQ4). The single-run ladder looks like a tidy climb ($0.804 \rightarrow 0.840$), but under out-of-fold masks, five-seed ensembling and bootstrap CIs the variants' intervals overlap almost entirely (Table 3, Fig. 1). Re-running the *identical* configuration with only the seed changed moves macro-F1 by up to 0.039 – comparable to the entire 0.036 "improvement" of the ladder; a study reporting the shape variant's luckiest seed against a no-feature unlucky one would "show" a +0.043 gain that is pure noise. Hand-crafted features give *no reliable* gain: *none* and *shape* both ensemble to 0.875. Only *ensembling* survives (per-seed $0.847 \rightarrow$ ensemble 0.875, above even the best individual seed). Our best defensible result is macro-F1 **0.875** (95% CI $[0.80, 0.94]$), malignant recall 0.903.

5.4 Error analysis

Classification confusion concentrates on the malignant class and the benign↔malignant boundary, matching the EDA finding that the two overlap in intensity. Segmentation errors track lesion difficulty – small low-contrast lesions and irregular malignant boundaries drive the large Dice std, and the worst cases are near-complete *misses*

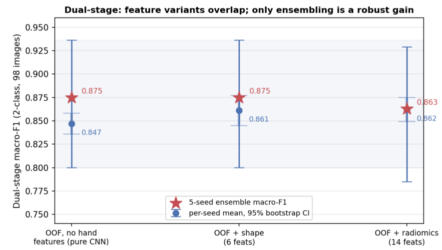

Figure 1: The three feature variants' bootstrap 95% CIs overlap almost entirely; only ensembling (stars) reliably lifts the per-seed mean (dots).

rather than uniform boundary error, exactly the subpopulation the pretrained encoder most improves. In the pipeline the residual errors are classification errors on correctly localised lesions.

6 Conclusion

Answers. *RQ1:* transfer-learned models lead each task and reproduced SOTA scores below published values. *RQ2:* the pretrained encoder is decisive (+0.117 Dice); loss, attention, TTA and nnU-Net add little. *RQ3:* better segmentation does *not* improve the pipeline – it is classifier-bound. *RQ4:* single-run gains are unstable; only ensembling survives.

Strengths. The dominant factor shaping our numbers is evaluation hygiene, and that discipline – remove leakage, attribute gains with controlled ablations, report multi-seed results with CIs – is the project's central contribution. That published scores fall sharply under a cluster-grouped split does not mean our implementations are weak; it means BUSI leaderboard numbers are incomparable unless splitting and de-duplication match. Measuring the bottleneck with a ground-truth upper bound *before* investing in the more visible component is a transferable lesson.

Limitations. (i) All results come from one small dataset and one split; 98 test images cannot resolve small differences, and the test-set uncertainty (±0.07) dwarfs the training uncertainty (±0.01), so no number of seeds would fix this. (ii) No external or multi-centre validation. (iii) BUSI itself has near-duplicate and annotation issues. (iv) We use static 2D frames and the data carry acquisition bias. (v) Compute limited our search: we ran no CLAHE/denoising ablation, a deliberate omission rather than evidence those steps are useless. (vi) Our reproductions may differ in detail from the original code, and the dual-stage task is two-class, so not comparable to the three-class classifiers. We therefore interpret the system as *decision support*: none of these numbers would justify unsupervised clinical use. **Given more time** we would add external multi-centre and nested cross-validation; audit the classifiers with Grad-CAM to confirm they attend to the lesion, not to burned-in annotations; and, since the pipeline is classifier-bound, invest in stronger ROI classifiers and self-supervised pretraining rather than in the segmenter.

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